

# Software Requirements Specification

*B.Tech CSE 2025–29 · Software Engineering & Project Management · Semester III*

Dashboard basis: The project author has developed three dashboards—Student, Faculty and Admin. This document uses the supplied walkthrough to describe them and distinguishes those dashboard details from requirements, integrations or metrics that are proposed or not independently verified.

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## 1. Purpose and overview

CourseCraft is a proposed online learning platform for a university continuing-education centre. It is intended to help working professionals discover and enrol in short courses, pay the stated course fee, study video lessons, complete a quiz and receive a certificate when they meet the course rules. Faculty need tools to manage course content and monitor learners; editors need a dependable production workflow; and the Director needs a clear view of readiness, enrolments and delivery progress.

This SRS combines the assignment's case facts with the supplied description of the three dashboards. It describes intended behaviour; it is not an independent audit of the implementation. Where the source material does not specify a policy, this document makes a small, explicit assumption and marks it for stakeholder confirmation.

## 2. Scope

### In scope

- A course catalogue for 30 short courses, with a staged initial release of up to 8 courses if the remaining content is not ready.
- Student registration, sign-in, enrolment and payment records at **₹4,500 per enrolment**.
- Video lessons, learner progress, a 10-question course quiz, and certificate eligibility.
- Faculty course, recording, question-bank and learner-progress views.
- An administrator dashboard for course readiness, production capacity, enrolment and revenue metrics.
- Role-based access for students, faculty and administrators.

### Out of scope

- The university's teaching decisions and faculty availability; the platform can display and record these, but cannot create recording capacity.
- Payment-gateway settlement or refund policy, unless separately approved and integrated.

- Creation of the actual course videos and academic content.
- Native mobile applications and live video conferencing.

### 3. Stakeholders and needs

| Stakeholder                      | Main need                                                                             | Platform response                                      |
|----------------------------------|---------------------------------------------------------------------------------------|--------------------------------------------------------|
| Learners / working professionals | Find a course, enrol, learn around work, see progress and earn a credible certificate | Student dashboard, course player, progress and results |
| Faculty                          | Organise course material, maintain quiz questions and understand learner progress     | Faculty dashboard and course-management functions      |
| Editors / production team        | Know what is planned, recorded, awaiting edit or ready                                | Production status and workload visibility              |
| Director / administrator         | See launch readiness, revenue assumptions and delivery risks                          | Admin dashboard and staged-launch metrics              |
| IT / operations                  | Keep access controlled and core services available                                    | Authentication, role checks, monitoring and backups    |

### 4. Assumptions and constraints

1. Each course contains **12 recorded-content hours**. The case's production factor is **3 hours of combined recording/editing effort per content hour**, so the planning effort is 36 hours per course.
2. The available editing capacity is **2 editors × 30 hours per week = 60 editor-hours per week**. This is a planning ceiling, not a guarantee of uninterrupted capacity.
3. The stated faculty constraint is that faculty can record only 8 courses in the next three months. The 4.8-week calculation for 8 courses is the editor-capacity estimate; it does **not** override faculty availability or prove that faculty will deliver on that schedule. The apparent mismatch is tracked as a risk.
4. The platform can be ready in 10 weeks, as stated in the case. Full production of 30 courses is estimated at 18 weeks, so a phased release is recommended.
5. The dashboard walkthrough describes an intended fee of ₹4,500, 30 catalogue entries, progress gating at 100%, a 10-question quiz with a 70% pass mark, and role-specific dashboards. These are treated as supplied design details, not independently verified facts.
6. Payment provider, data-retention period, certificate-verification policy, service-level owner and legally approved privacy wording have not been specified and require confirmation before production use.

### 5. Functional requirements

MoSCoW: **Must** = essential for the first usable release; **Should** = important but a controlled workaround is possible; **Could** = useful enhancement; **Won't (this release)** = explicitly deferred.

| ID    | Requirement                                                                                                                      | Priority | Acceptance evidence                                                                                     |
|-------|----------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------|
| FR-01 | The system shall display the available courses with course code, title, faculty, duration, modules, availability and fee.        | Must     | TC-01 catalogue displays the published course records and ₹4,500 fee.                                   |
| FR-02 | A learner shall be able to create an account and sign in using valid credentials.                                                | Must     | TC-02 successful registration/login; TC-03 invalid credentials are rejected.                            |
| FR-03 | The system shall permit a signed-in learner to start enrolment in an available course and create a transaction/enrolment record. | Must     | TC-04 enrolment record is associated with the learner and course.                                       |
| FR-04 | The system shall record the course fee as ₹4,500 per enrolment and retain a transaction reference and payment status.            | Must     | TC-05 fee and transaction fields match the selected course; failures do not mark payment as successful. |
| FR-05 | An enrolled learner shall be able to view authorised course lessons and play the course video.                                   | Must     | TC-06 enrolled learner can play; unauthorised learner is denied.                                        |
| FR-06 | The system shall record lesson completion and calculate course progress from completed lessons.                                  | Must     | TC-07 progress is recalculated and displayed consistently.                                              |
| FR-07 | The system shall unlock the certification quiz only when course progress is 100%.                                                | Must     | TC-08 99% remains locked; TC-09 100% unlocks.                                                           |
| FR-08 | The system shall present a 10-question multiple-choice assessment and calculate the result as correct answers divided by 10.     | Must     | TC-10 question count and score calculation are correct.                                                 |
| FR-09 | The system shall mark a learner as passed when the score is at least 70%, and not passed below 70%.                              | Must     | TC-11 6/10 fails; TC-12 7/10 passes.                                                                    |
| FR-10 | The system shall issue and store a certificate only for an eligible learner, and let that learner view or download it.           | Must     | TC-13 eligible pass issues one certificate; ineligible attempts issue none.                             |
| FR-11 | Faculty shall be able to view their assigned courses, update permitted quiz questions and review learner progress and results.   | Should   | TC-14 faculty actions are limited to assigned courses and permitted fields.                             |
| FR-12 | Administrators shall be able to review course readiness, production status, enrolment and revenue metrics across the catalogue.  | Should   | TC-15 administrator view aggregates all authorised records and identifies assumptions/targets.          |

## 6. Non-functional requirements

The targets below are measurable proposed acceptance criteria. They should be confirmed against the university's hosting budget and operational needs before production release.

| ID     | Quality attribute            | Measurable requirement / acceptance test                                                                                                                                                                | Priority |
|--------|------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| NFR-01 | Video availability           | Course-video playback service shall achieve at least 99.5% monthly availability, excluding announced maintenance; measure using 5-minute synthetic checks.                                              | Must     |
| NFR-02 | Video start performance      | For supported broadband connections (at least 10 Mbps), the 95th-percentile time from Play to first frame shall be $\leq 3$ seconds over a monthly sample of at least 1,000 starts.                     | Should   |
| NFR-03 | Security in transit          | All production login, enrolment, payment-status and certificate traffic shall use HTTPS/TLS 1.2 or later; automated deployment check shall find no HTTP endpoint for these flows.                       | Must     |
| NFR-04 | Role and record access       | In a suite of at least 30 role-based negative-access tests, no student shall read faculty/admin data or another student's private record; all unauthorised requests shall return 401/403 and be logged. | Must     |
| NFR-05 | Credential protection        | Passwords shall never be stored or logged in plaintext; storage shall use a salted adaptive hash (Argon2id or bcrypt with work factor $\geq 12$ ); verify by configuration and code review.             | Must     |
| NFR-06 | Scalability                  | The service shall support 500 concurrent signed-in users with 95th-percentile API response time $\leq 2$ seconds for non-video API requests and error rate $< 1\%$ during a 30-minute load test.        | Should   |
| NFR-07 | Data durability and recovery | Enrolments, payment references, progress and certificates shall be backed up at least every 24 hours; a quarterly restore drill shall demonstrate RPO $\leq 24$ hours and RTO $\leq 4$ hours.           | Should   |

| ID     | Quality attribute               | Measurable requirement / acceptance test                                                                                                                                 | Priority |
|--------|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| NFR-08 | Accessibility and compatibility | Core learner flows shall meet WCAG 2.1 AA on automated checks plus manual keyboard review, and pass current stable Chrome, Firefox, Edge and Safari desktop smoke tests. | Should   |

## 7. External interfaces and data

- **User interface:** Responsive web pages for public course discovery, student learning, faculty work and administration.
- **Backend interface:** HTTPS REST API for identity, courses, enrolments, progress, quiz results, certificates and dashboards.
- **Video delivery:** A CDN or equivalent video host serves media separately from application APIs. This keeps high-bandwidth media transfer away from the application server.
- **Payment interface:** The supplied walkthrough mentions a checkout request, but it does not identify a live payment provider or settlement verification. A production integration must use provider-confirmed status and must never treat a client-side success screen as payment evidence.
- **Persistence:** The walkthrough describes Firestore collections and local JSON persistence. For production, one authoritative transactional store and a tested backup/restore strategy must be selected; dual writes need reconciliation and failure handling.
- **Core records:** User (role and profile), course, lesson, enrolment, payment reference/status, progress, quiz attempt, certificate and production status.

## 8. Business rules

7. The course fee is ₹4,500 per enrolment for the case calculations.
8. A learner must be enrolled and reach 100% course completion before taking the certificate quiz.
9. The assessment contains 10 questions; score percentage =  $\text{round}(\text{correct answers} \div 10 \times 100)$ .
10. Score  $\geq 70\%$  means passed; score  $< 70\%$  means not passed.
11. A certificate may be issued only when both completion and pass conditions are true. Repeated submissions must not create duplicate certificates for the same course completion.
12. A faculty user may manage only courses assigned to that user; a student may access only their own private learning records.

## 9. Traceability matrix

| Requirement | Test case(s) |
|-------------|--------------|
| FR-01       | TC-01        |
| FR-02       | TC-02, TC-03 |
| FR-03       | TC-04        |

| Requirement | Test case(s) |
|-------------|--------------|
| FR-04       | TC-05        |
| FR-05       | TC-06        |
| FR-06       | TC-07        |
| FR-07       | TC-08, TC-09 |
| FR-08       | TC-10        |
| FR-09       | TC-11, TC-12 |
| FR-10       | TC-13        |
| FR-11       | TC-14        |
| FR-12       | TC-15        |

## 10. Open decisions before baseline approval

- Confirm payment provider, callback/signature validation and refund/cancellation rules.
- Confirm privacy notice, retention and deletion rules, including student records and video analytics.
- Confirm whether certificates require a public verification page or revocation mechanism.
- Confirm the actual video host/CDN and service-level owner.
- Reconcile the 8-course faculty commitment with the 60 editor-hours/week capacity estimate.
- Replace proposed NFR targets with approved service objectives if university policy requires different values.